

## MTH204 (Section A)

### Quiz-1

February 11, 2026

1 hour and 15 minutes

Total Marks: 30 (Each question carries 5 marks)

Name:

Roll No.:

**Important Notice:** Only a pen and a single **A4-sized handwritten cheat sheet** are permitted during the quiz. No other materials, devices, or aids are allowed. **No discussion or communication of any kind is permitted.** Any violation of these rules will result in the award of **zero marks** for the quiz, and the incident will be formally reported.

**Q1.**

In a fish farm, fish are harvested at a constant rate of 2100 fish per week. The per-capita birth rate for the fish is 0.7 fish per day per fish and the per-capita death rate is 0.2 fish per day per fish.

1. Write down a word equation describing the rate of change of the fish population. Hence obtain a differential equation for the number of fish. (Define any symbols you introduce.)
2. Find the general solution of the differential equation obtained in part (a).
3. Determine if there are any values for which the fish population is in equilibrium.

**Q2.** Let  $y_1$  and  $y_2$  be any two solutions of the second order differential equation

$$a_0(x) y'' + a_1(x) y' + a_2(x) y = 0 \quad \forall x \in [a, b]$$

where  $a_0(x) \neq 0$  for all  $x \in [a, b]$  and  $a_0(x)$ ,  $a_1(x)$  and  $a_2(x)$  are continuous for all  $x \in [a, b]$ . Then show that the Wronskian  $W$

$$W(x) = C e^{-\int \frac{a_1(x)}{a_0(x)} dx}$$

where  $C$  is any constant.

**Q3.** Find the general solution of the differential equation

$$y'' - 3y' + 2y = 2x^2 + e^x + 2xe^x + 4e^{3x}.$$

**Q4.** If the differential equation

$$\frac{dy}{dx} = \sqrt{x^2 + y^2} \quad y(1) = 2$$

is solved using Euler's Method with step size  $h = 0.1$ , then find the value of  $y(1.2)$  (round off to 2 places of decimals). **USE:**  $\sqrt{5} = 2.23606$ ,  $2.22360^2 = 4.944396$  **and**  $\sqrt{6.15439696} = 2.480805708..$

**Q5.** Consider the non- homogeneous differential equation

$$(\sin^2 x) y'' - (2 \sin x \cos x) y' + (\cos^2 x + 1) y = \sin^3 x.$$

on  $[\pi/4, 3\pi/4]$ . Check that  $y_1 = \sin x$  and  $y_2 = x \sin x$  are solutions of the corresponding homogeneous equation, show that they're linearly independent on the given interval and find the general solution.

**Q6.** For what values of  $a$  in  $\mathbb{R}$  the first-order differential equation

$$\left| \frac{dy}{dx} \right| + |y| + a = 0$$

has the real solution.